SDC HACKATHON 2026

Product Requirement Document (Updated for the Final Round)

Future Skills · Team Hands Off Our Rights · Domain 1: Aiversecity Future-Ready Workforce

Revision of "SDC2026_HandsOffOurRights_Preliminary.pdf" (25 October 2026). Updated 1 November 2026.

0. What Changed Since the Preliminary Submission, and Why

The Final Round Finalist Guide raised the bar: all six must-haves must run live, judges may change the trend data mid-demo, check that a course is real, and ask why a score is what it is. What we learned from the data and from testing the flow shaped the rest.

Area	Preliminary (25 Oct)	Final (1 Nov)	Why
IT areas and roles	6 areas, 26 roles	5 areas, 25 roles: IT Management dropped; Statisticians and Special Effects Artists and Animators added	IT Management roles had no usable Malaysian demand signal; the two added roles widen the data and creative options for students
Trend score	Mostly DOSM online job ads	v3: Malaysian demand = share of an area's roles on MyCOL 2024/2025 (+ DOSM where it exists); AI momentum = continuous arXiv growth scale	DOSM publishes job ads for only one IT occupation, so the old score barely separated areas; MyCOL is official and per-occupation
Change test	Update data button only	"What if the trend data changed?" page: change any input and see signals → scores → ranking before and after	Judges will ask "change the trend data now, what changes?"
Trend radar	One page among many	Landing page, with source and date links for every input, and Update data at the top	Must-have 2 requires source and date on screen
Learning pathway	54 of 186 needed skills (29.0%) had a free checked course or video; 26 (14%) counting courses only	166 of 191 (86.9%), 83 (43.5%) from courses alone; every job has its own resources; link checker; progress tracking without accounts	Phase 1 KPI 2 target was 40%; judges may check a course is real
Matches page	Long cards	Scenario switch on top, SDG icons and score bars on every card, short AI "Why this job?"; a "Global goals you care about" picker: jobs that support a picked UN goal get 5 extra points and move up	The scenario and the score parts should be seen first; the guide says SDG tags must change something for the user
Languages and access	English	English, Bahasa Melayu, Chinese; light/dark mode; gzip and ETags; 40 px touch targets	Good-to-have "inclusive design" and low-bandwidth use
LLM	gpt-oss-120b on Fireworks AI (team's own account)	gpt-5.4-mini through the SDC sponsor KrackedDevs gateway; any OpenAI-compatible API works; same guardrails	Use the sponsor resource and avoid single-provider risk
Trust	—	Score ranges on the skills test, exam-section links per question, CV privacy box, progress bar while the CV is read	A 3-question area score is uncertain and a CV is personal data, so the app says so plainly
Positioning	—	Trends page card naming JobStreet, Hiredly, LinkedIn Jobs, MyFutureJobs, MyMahir, MyCOL, Coursera, Udemy, edX and yearly reports, and what each lacks for a student	The guide asks teams to name the job sites and course platforms that exist and show what makes theirs different

1. Team & Project Overview

Team Name	Hands Off Our Rights
Selected Domain	Domain 1: Aiversecity Future-Ready Workforce
SDG Track	SDG 4 Quality Education (4.3, 4.4); SDG 8 Decent Work and Economic Growth (8.2, 8.5)
Project / Solution Name	Future Skills

Name (Leader)	Hong Zhi Yang
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University Affiliation	Universiti Tunku Abdul Rahman (UTAR)
Number of Team Members	3
Live Prototype	https://future-skills.replit.app
Code Repository	https://github.com/pekoremon/future-skills

2. Problem Statement

2.1 Problem context. Malaysia's graduates often end up in jobs that do not use their education: in Q2 2026, 35.2% of employed people with tertiary education (1.95 million) worked in semi-skilled or low-skilled jobs (DOSM, 2026b). Youth unemployment (15–24) was 10.3% in 2024, against 3.2% nationally (DOSM, 2026a). AI is spreading faster than training: 56% of Malaysian professionals use AI tools at work but only 26% have had formal training (Hays Asia Salary Guide 2025, quoted in the Domain 1 brief).

We focus on IT and computing undergraduates choosing a specialisation, an internship or a first job. "IT" is really five different areas (software engineering, data and AI, cybersecurity, cloud and networking, web and interface design), each needing different interests and skills, and AI is changing which roles are in demand faster than course guides are updated. The information exists (TalentCorp MyCOL, DOSM job-ad and wage data, free courses) but is scattered and not linked to the individual. No tool connects it to one student's interests and skills, or turns a skills gap into a plan.

2.2 Target users. Primary: IT and computing undergraduates in Malaysian universities, polytechnics and TVET colleges. Secondary: career centres advising them, and working adults reskilling into IT (HRD Corp-claimable courses are listed for them).

3. Solution and Key Features

Future Skills is a web app that shows an IT student which IT area suits them, which of 25 IT roles fit, which skills they lack, and a free plan to learn them, with Malaysian demand and AI-trend signals for every role. **GenAI explains; our data decides.**

Must-have (Domain 1)	**What the student sees in the final build**
1. Skills profile intake	Optional CV (PDF/DOCX/text) or 1–5 skill ratings → 36-item interest questionnaire (RIASEC + IT area) → skills test on only the 3–5 areas these point to (15–21 questions, hands-on SQL and Python), with score ranges
2. AI-trend signal layer	Future Skills Radar landing page: MyCOL, DOSM/OpenDOSM and arXiv inputs with source and date, direction and confidence per area and role; "Update data" refreshes live; "What if" lets anyone change an input
3. Future-role matching	25 roles ranked by 0.4 area fit + 0.1 interest + 0.3 skill + 0.2 trend; every part shown, plus the missing skills and what would raise the score
4. Personalised reskilling pathway	Gaps ordered biggest first, checked free courses and videos with hours and weeks at the student's study time, HRD Corp-claimable options listed apart, certificates and project ideas, progress tracking
5. SDG mapping	Every role and skill tagged to SDG targets with a one-line reason (233 links); the path sums up which targets it contributes to; the UN goals a student picks reorder their jobs (+5 points per supported job)
6. Handling uncertainty	Three scenarios (current trends, fast AI adoption, Malaysian demand leads) re-weight measured trend parts; rankings shift and are labelled as estimates

How AI is applied. (1) CV reading: an LLM (gpt-5.4-mini via KrackedDevs) quotes CV evidence per IT area; our code keeps only quotes really in the CV and scores them. The CV is never stored. (2) "Why this job?": the LLM turns one match's computed facts into plain language; a reply with any number not in those facts is rejected and a rule-based summary shown. (3) Matching, trends, scenarios and pathways are explainable code over sourced tables.

4. Scope and Boundaries

In scope	Out of scope
IT career choice for students: 5 IT areas, 25 roles, Malaysian demand data, free learning paths, SDG links, scenarios; English, BM and Chinese; phone and desktop	Non-IT careers; writing course content; payments; real recruitment integrations or hiring decisions (the student decides); storing CVs or personal data; real CVs in demos

5. Technical Architecture

Layer	Technology
Frontend	HTML, CSS, plain JavaScript (no framework, no build step); PDF.js and Pyodide in the browser; EN/BM/中文
Backend	Python 3.12 standard library: JSON API on http.server, gzip + ETags; no third-party packages
Data	CSV/JSON tables (roles, skills, role–skill links, courses, resources, SDG map, trend inputs/signals/scores); every row has source name, URL and date; validate_data.py and check_links.py
GenAI	Any OpenAI-compatible chat API; live: gpt-5.4-mini via the SDC sponsor KrackedDevs gateway, with output checks and rule-based fallback
External data APIs	OpenDOSM API, arXiv API (live refresh); Microsoft Learn Catalog API (development only)
Hosting / Dev	Replit Workspace and Deployments (SDC sponsor), live at https://future-skills.replit.app; GitHub; KrackedDevs GenAI keys (SDC sponsor)
Testing	216 unittest tests, data validator, link checker, fairness check, kpi.py

6. SDG Impact Framework and KPIs

Item	Phase 1 definition	Status at the Final (1 Nov 2026)
Target SDG sub-targets	4.4 relevant skills for decent jobs; supporting 8.5 full and productive employment	Unchanged; 4.3 and 8.2 also tagged where a role or skill supports them
Problem baseline	35.2% of tertiary-educated workers in semi/low-skilled jobs (DOSM Q2 2026); 26 of 186 skills (14%) with a checked course	Same national baseline; prototype baseline re-measured with scripts/kpi.py at the Phase 1 commit 6802b63: 54 of 186 (29.0%) counting free videos too
KPI 1: Career clarity	≥70% of ≥20 IT student testers can name a target role, its top missing skills and a next course after one session	Pilot at the physical hackathon: **20 testers: 35% before → 75% after (7 → 15 of 20); confidence 3.3 → 3.7 of 5**. Form: https://docs.google.com/forms/d/e/1FAIpQLSemWTTx0Gzn7CJ4igIBwnOeNsCOR4qJPu7kVO80mszVF2fy4A/viewform Responses: https://docs.google.com/spreadsheets/d/1ZorKD7hgimQIG9NURNKgt8dAjFOnagovdUyOE2hS7q4/edit
KPI 2: Pathway coverage	Raise checked free-course coverage to 40% (about 75 skills)	86.9% (166 of 191 needed skills; 43.5% from courses alone), measured by python scripts/kpi.py: target exceeded

7. Design Thinking, Deployment and Next Steps

Empathise / Define: we are IT students facing the same choice; students need roles they had not considered, proof that a trend is real, and a concrete next step. **Ideate:** transparent scoring over a black box, and one rule: never invent a number. **Prototype and test:** students gave up on the full 24-question test, so the flow became CV → interests → a focused test; the radar moved to the landing page because judges and students both start with "what is changing?". **Deploy:** the app is live on Replit, needs no packages, and still works without the LLM key or when a data source fails (the last good data is kept). **Next:** run the career-clarity study with partner career centres, add each new DOSM Job Market Insights quarter and MyCOL edition, skill-level signals from job ads, and an HR team view. Done: a fairness check (the same CV as 8 different students gives identical scores).

8. Sources

• DOSM (2026a). Labour Market Review, Q1 2026. https://storage.dosm.gov.my/analysis/lmr_2026-q1_en.pdf

• DOSM (2026b). Labour Market Review, Q2 2026. https://storage.dosm.gov.my/analysis/lmr_2026-q2_en.pdf

• TalentCorp (2025). Malaysia Critical Occupations List (MyCOL) 2024/2025. https://www.talentcorp.com.my/mycol/

• Hays (2025). Hays Asia Salary Guide 2025, as quoted in the SDC 2026 Domain 1 brief.

• United Nations. Global indicator framework for the SDGs (A/RES/71/313). https://unstats.un.org/sdgs/indicators/indicators-list/

• Future Skills team (2026). Data tables and SOURCES.md. https://github.com/pekoremon/future-skills